UNIVERSIDAD CATOLICA BOLIVIANA ”SAN PABLO”

UNIDAD ACADÉMICA REGIONAL LA PAZ

INGENIERÍA DE TELECOMUNICACIONES

Laboratorio PRÁCTICA INTEGRADORA DE LAS UNIDADES 1 Y 2·

NOMBRE Carlos Andres Fernandez Guzman

ETAPA 0 — Preparación y resguardo

servidor

Last login: Sun Oct 4 07:36:59 2026 from 10.0.2.15

carlos-ucb-2@telecomunicaciones-tel332:~$ ip -br addr

lo UNKNOWN 127.0.0.1/8 ::1/128

enp0s3 UP 10.0.2.8/24 metric 100 fe80::a00:27ff:fe2a:3125/64

enp0s8 UP 192.168.50.11/27 fe80::a00:27ff:fe1c:1eda/64

carlos-ucb-2@telecomunicaciones-tel332:~$ ip route

default via 10.0.2.1 dev enp0s3 proto dhcp src 10.0.2.8 metric 100

8.8.4.4 via 10.0.2.1 dev enp0s3 proto dhcp src 10.0.2.8 metric 100

8.8.8.8 via 10.0.2.1 dev enp0s3 proto dhcp src 10.0.2.8 metric 100

10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.8 metric 100

10.0.2.1 dev enp0s3 proto dhcp scope link src 10.0.2.8 metric 100

192.168.50.0/27 dev enp0s8 proto kernel scope link src 192.168.50.11

carlos-ucb-2@telecomunicaciones-tel332:~$

cliente

carlos-ucb-1@telecomunicaciones-tel331:~$ ip -br addr

lo UNKNOWN 127.0.0.1/8 ::1/128

enp0s3 UP 10.0.2.6/24 metric 100 fe80::a00:27ff:fea7:d0c5/64

enp0s8 UP 192.168.50.25/27 metric 100 fe80::a00:27ff:fe0f:9efc/64

carlos-ucb-1@telecomunicaciones-tel331:~$ ip route

default via 10.0.2.1 dev enp0s3 proto dhcp src 10.0.2.6 metric 100

default via 192.168.50.11 dev enp0s8 proto dhcp src 192.168.50.25 metric 100

8.8.4.4 via 10.0.2.1 dev enp0s3 proto dhcp src 10.0.2.6 metric 100

8.8.8.8 via 10.0.2.1 dev enp0s3 proto dhcp src 10.0.2.6 metric 100

10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.6 metric 100

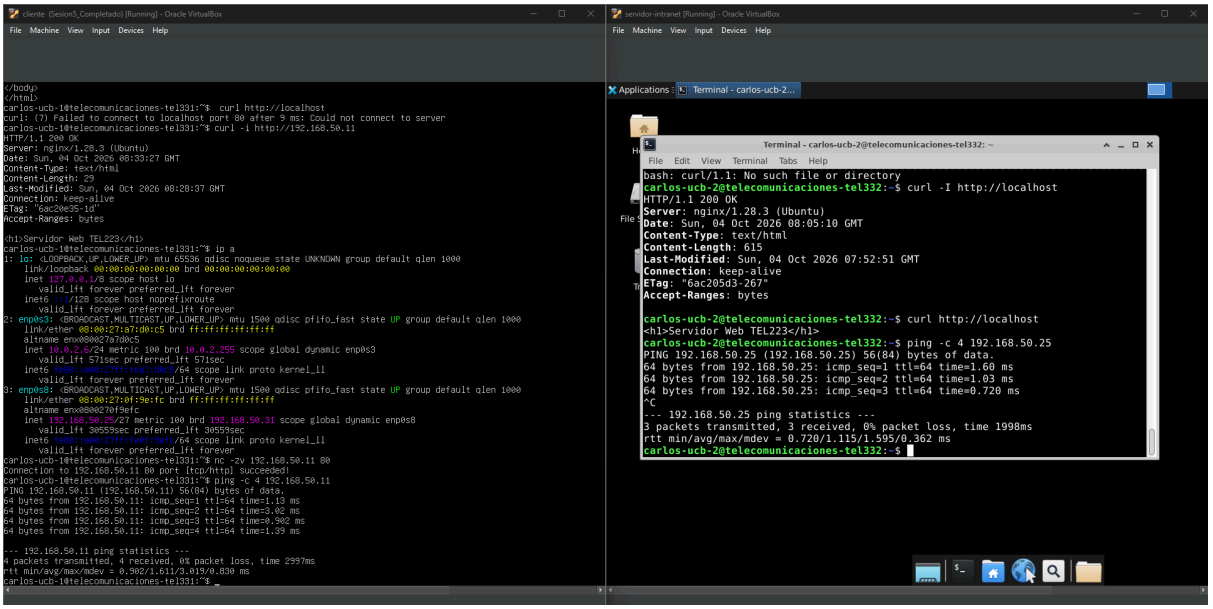
10.0.2.1 dev enp0s3 proto dhcp scope link src 10.0.2.6 metric 100

192.168.50.0/27 dev enp0s8 proto kernel scope link src 192.168.50.25 metric 100

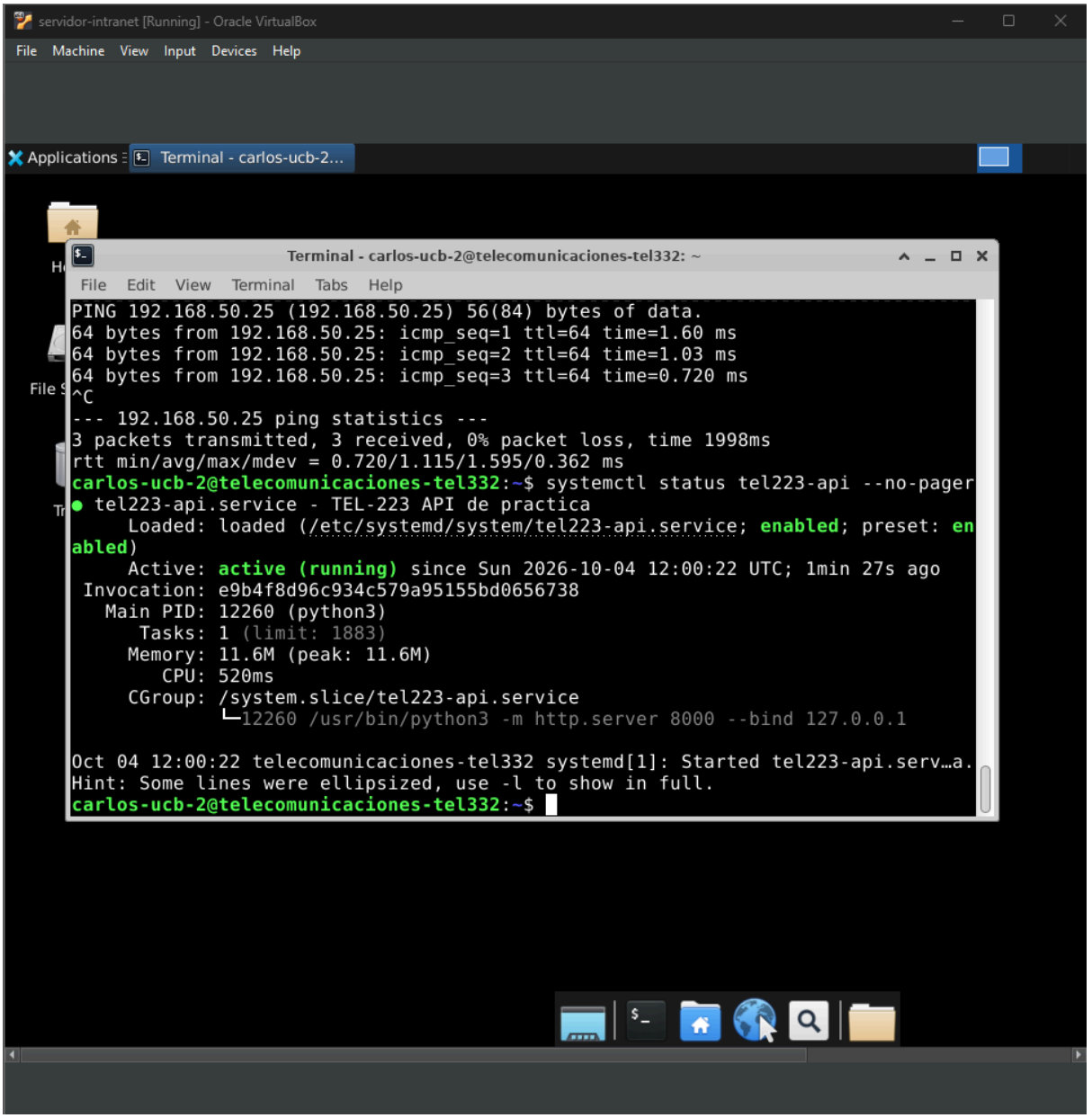
192.168.50.11 dev enp0s8 proto dhcp scope link src 192.168.50.25 metric 100

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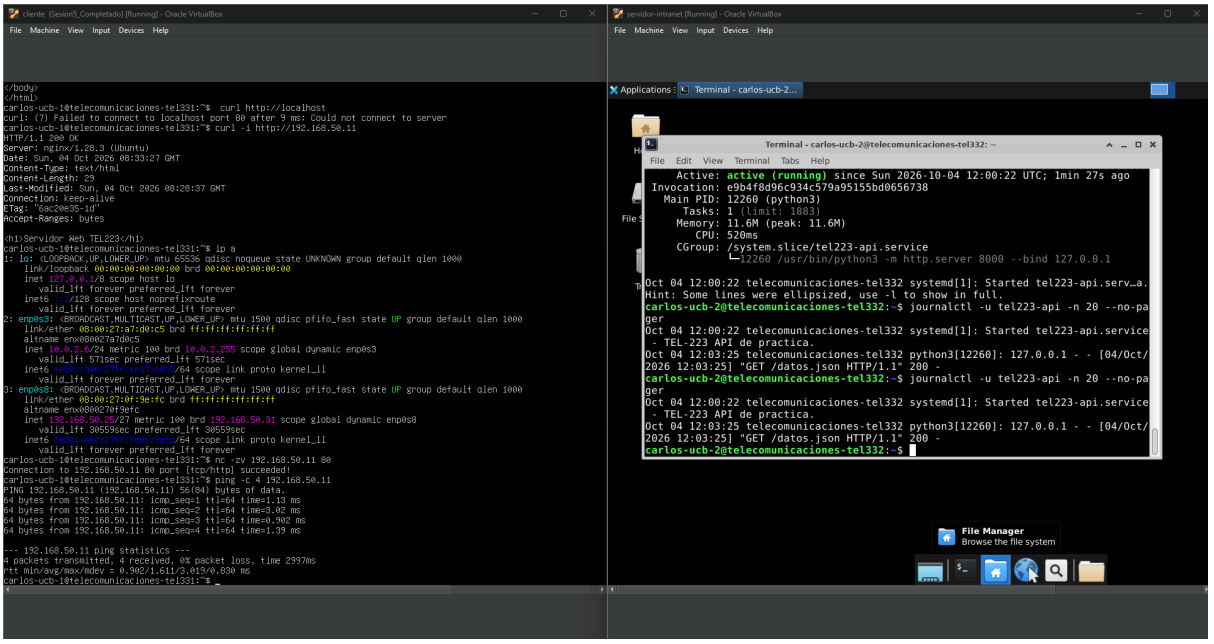
TEL-223 · Servicios de Intranet Internet



ETAPA 1 — Usuarios, permisos, servicios y registros


servidor-intranet [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Applications
Terminal - carlos-ucb-2...
Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
PING 192.168.50.25 (192.168.50.25) 56(84) bytes of data.
64 bytes from 192.168.50.25: icmp_seq=1 ttl=64 time=1.60 ms
64 bytes from 192.168.50.25: icmp_seq=2 ttl=64 time=1.03 ms
64 bytes from 192.168.50.25: icmp_seq=3 ttl=64 time=0.720 ms
^C
--- 192.168.50.25 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 0.720/1.115/1.595/0.362 ms
carlos-ucb-2@telecomunicaciones-tel332:~$ systemctl status tel223-api --no-pager
● tel223-api.service - TEL-223 API de practica
Loaded: loaded (/etc/systemd/system/tel223-api.service; enabled; preset: enabled)
Active: active (running) since Sun 2026-10-04 12:00:22 UTC; 1min 27s ago
Invocation: e9b4f8d96c934c579a95155bd0656738
Main PID: 12260 (python3)
Tasks: 1 (limit: 1883)
Memory: 11.6M (peak: 11.6M)
CPU: 520ms
CGroup: /system.slice/tel223-api.service
└─12260 /usr/bin/python3 -m http.server 8000 --bind 127.0.0.1
Oct 04 12:00:22 telecomunicaciones-tel332 systemd[1]: Started tel223-api.serv…a.
Hint: Some lines were ellipsized, use -l to show in full.
carlos-ucb-2@telecomunicaciones-tel332:~$

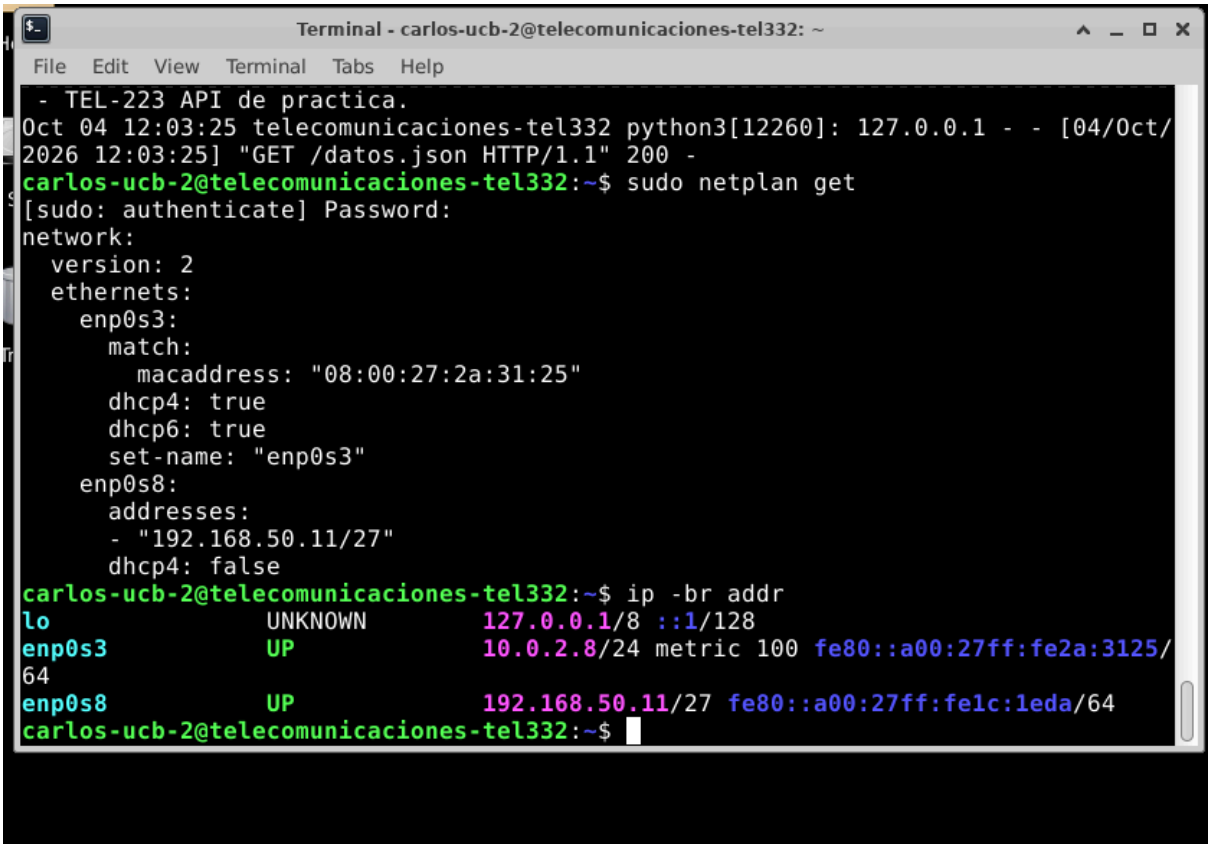
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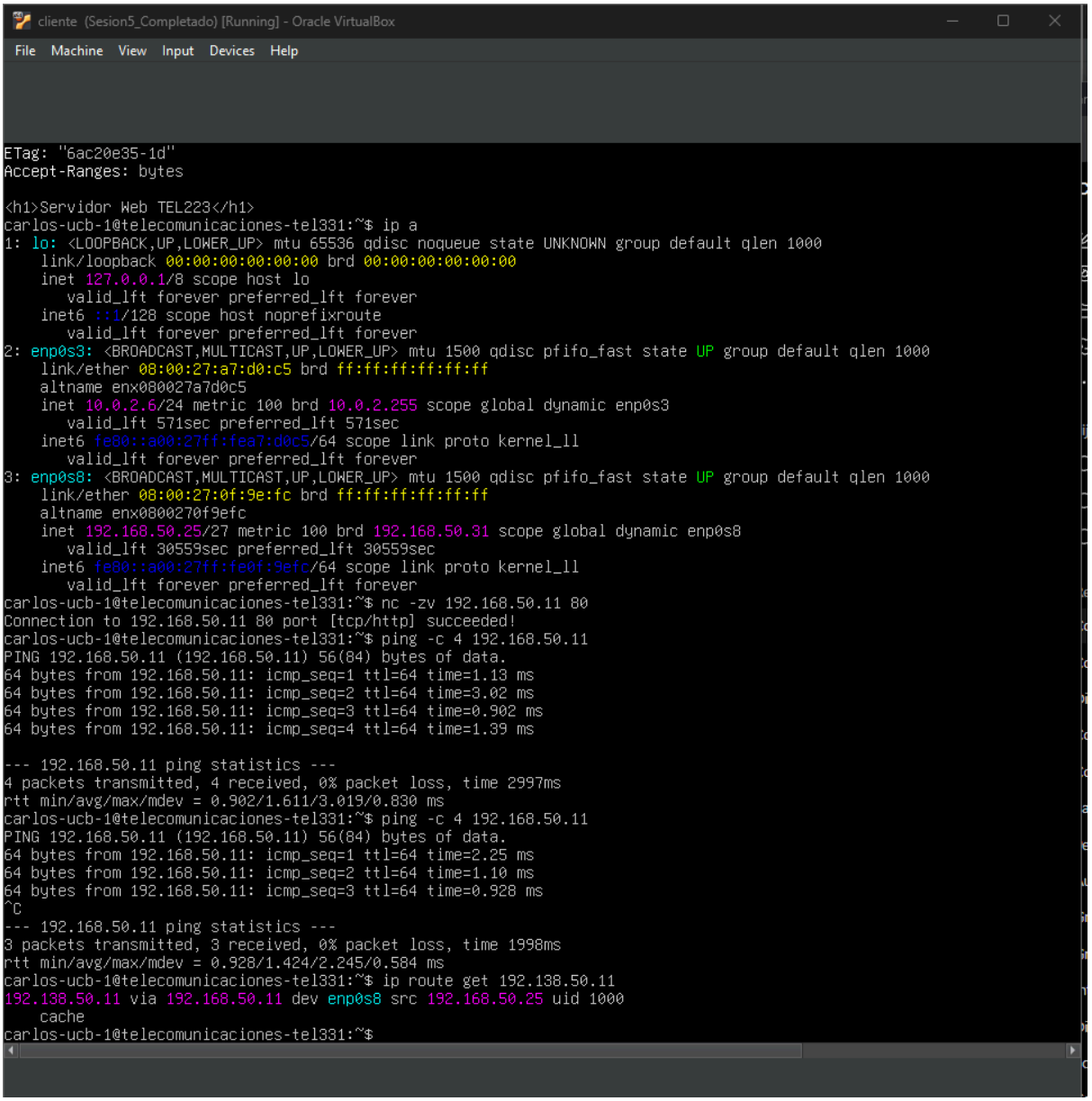


donde u/s	codigo
servidor	sudo useradd --system --no-create-home --shell /usr/sbin/nologin tel223
servidor	sudo mkdir -p /srv/tel223/api
servidor	echo '{"servicio":"tel223-api", "estado": "ok"}' | sudo tee /srv/tel223/api/datos.json
servidor	sudo chown -R tel223:tel223 /srv/tel223
servidor	sudo chmod 750 /srv/tel223/api
servidor	sudo chmod 640 /srv/tel223/api/datos.json
servidor	ls -l /srv/tel223/api
servidor	sudo ls -l /srv/tel223/api
servidor	sudo ls -ld /srv/tel223 /srv/tel223/api
servidor	systemctl status tel223-api --no-pager
servidor	sudo nano /etc/systemd/system/tel223-api.service
servidor	[Unit] Description=TEL-223 API de practica After=network.target [Service] User=tel223 WorkingDirectory=/srv/tel223/api ExecStart=/usr/bin/python3 -m http.server 8000 --bind 127.0.0.1 Restart=on-failure [Install] WantedBy=multi-user.target
servidor	sudo ls -l /etc/systemd/system/tel223-api.service

servidor	sudo systemctl daemon-reload
servidor	sudo systemctl enable --now tel223-api
servidor	si aparece run
servidor	sudo ss -ltnp | grep 8000
servidor	curl http://127.0.0.1:8000/datos.json

ETAPA 2 — Direccionamiento y conectividad

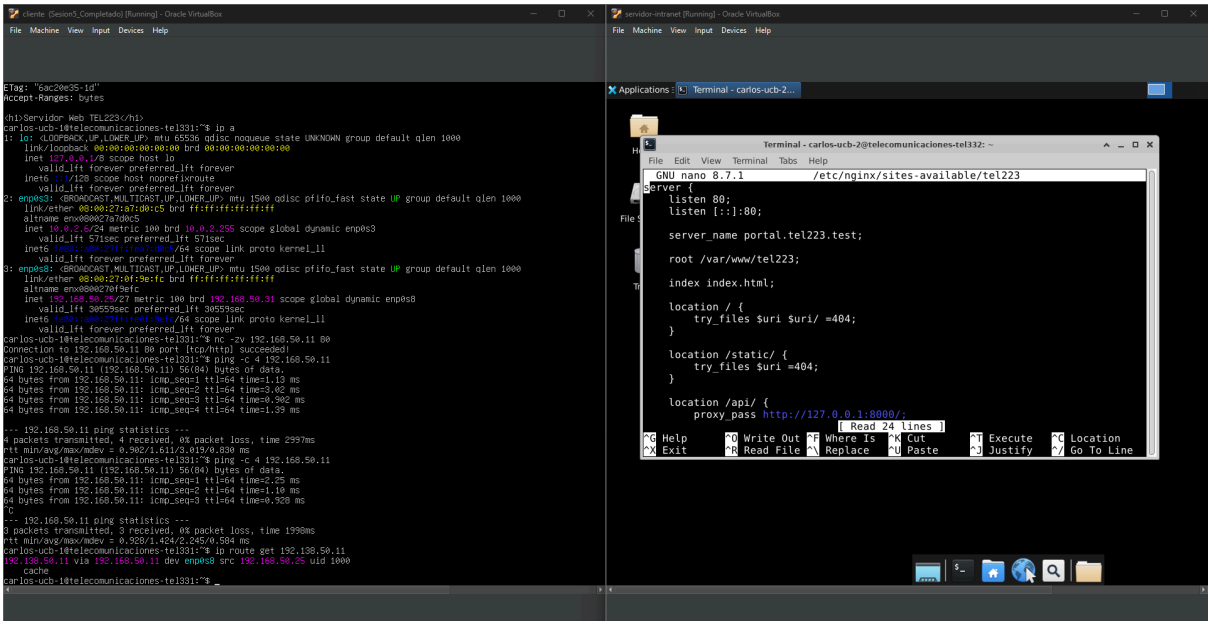


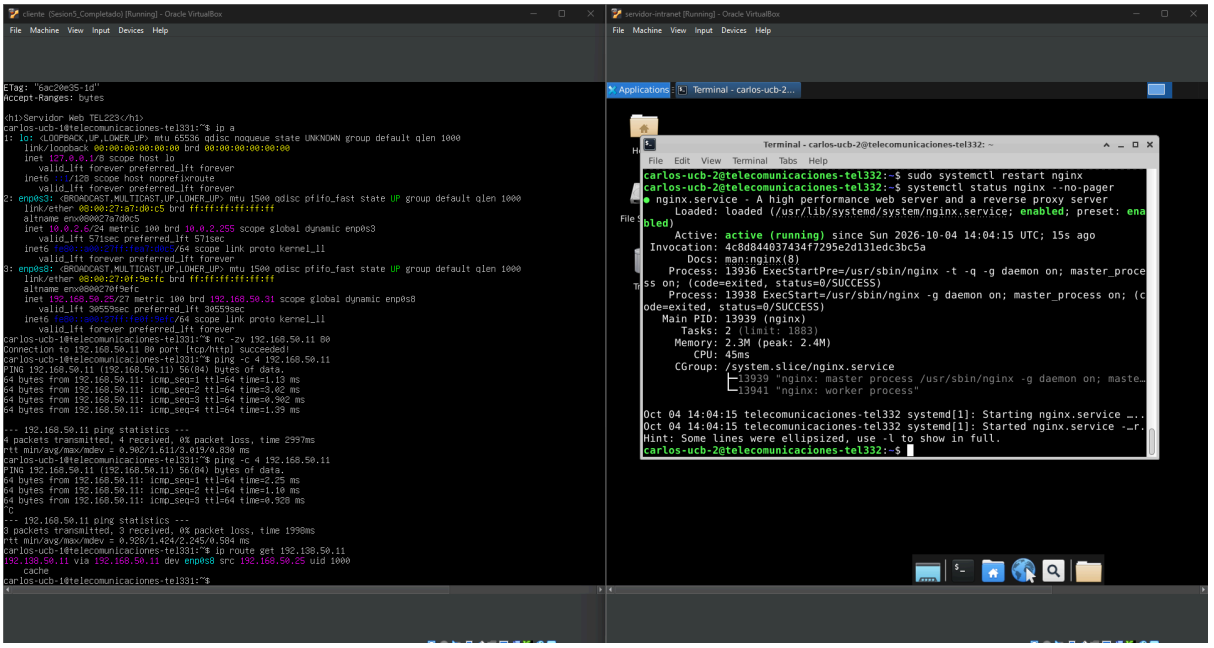


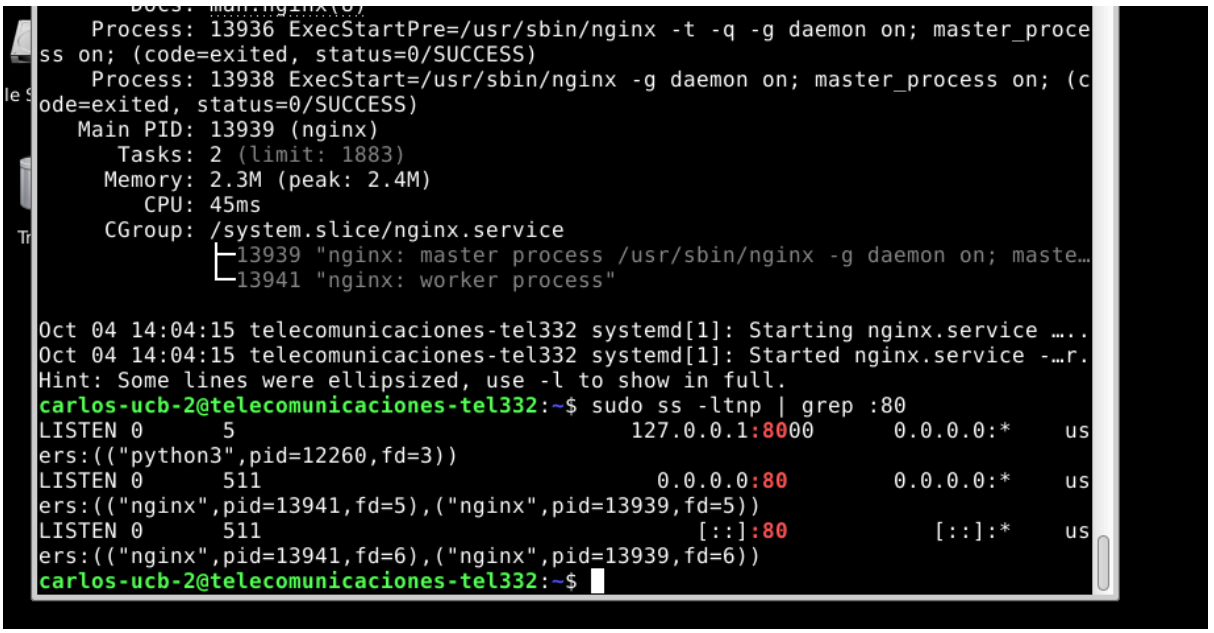
ETAPA 3 — DNS y DHCP con dnsmasq

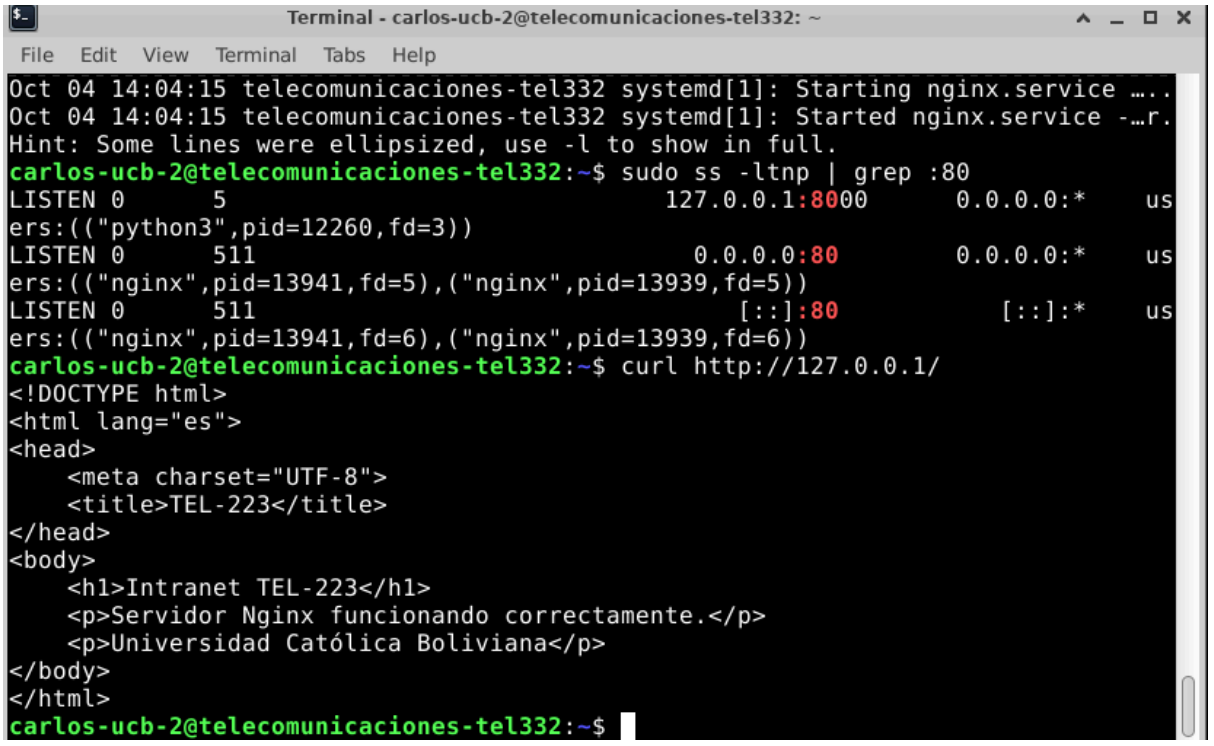
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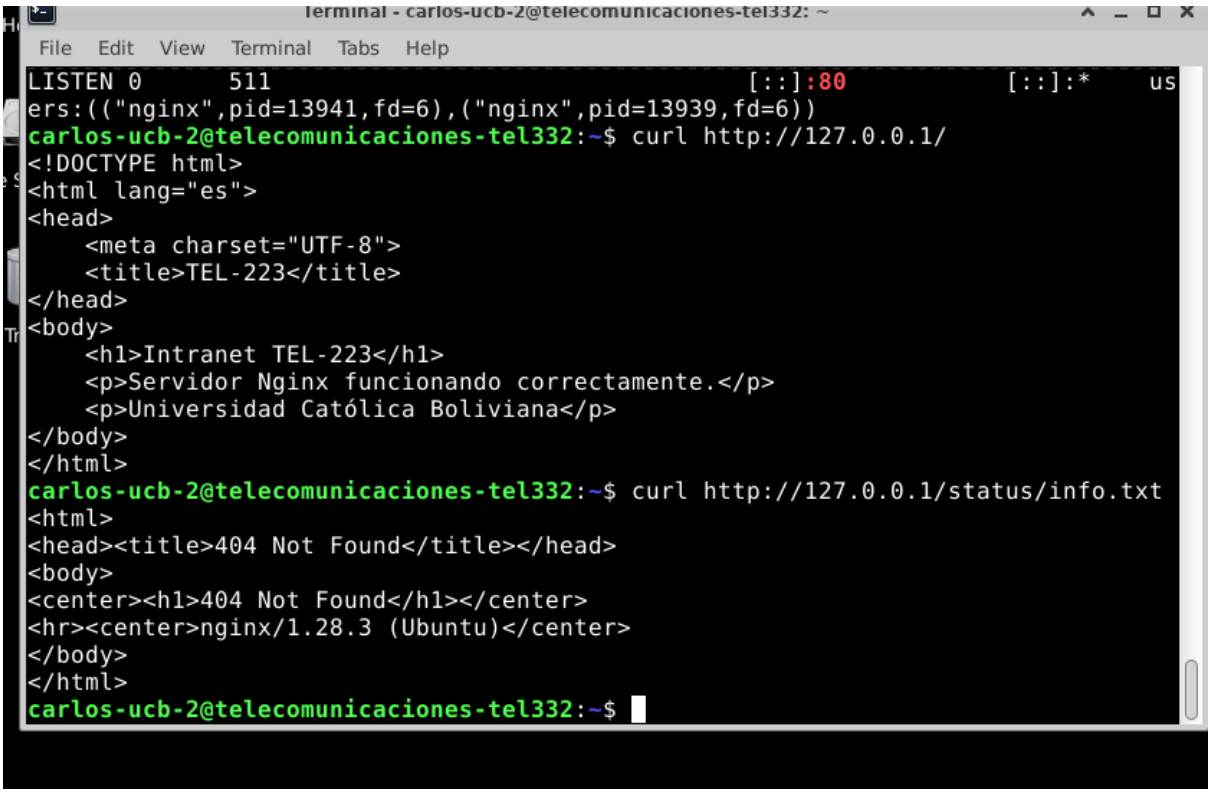
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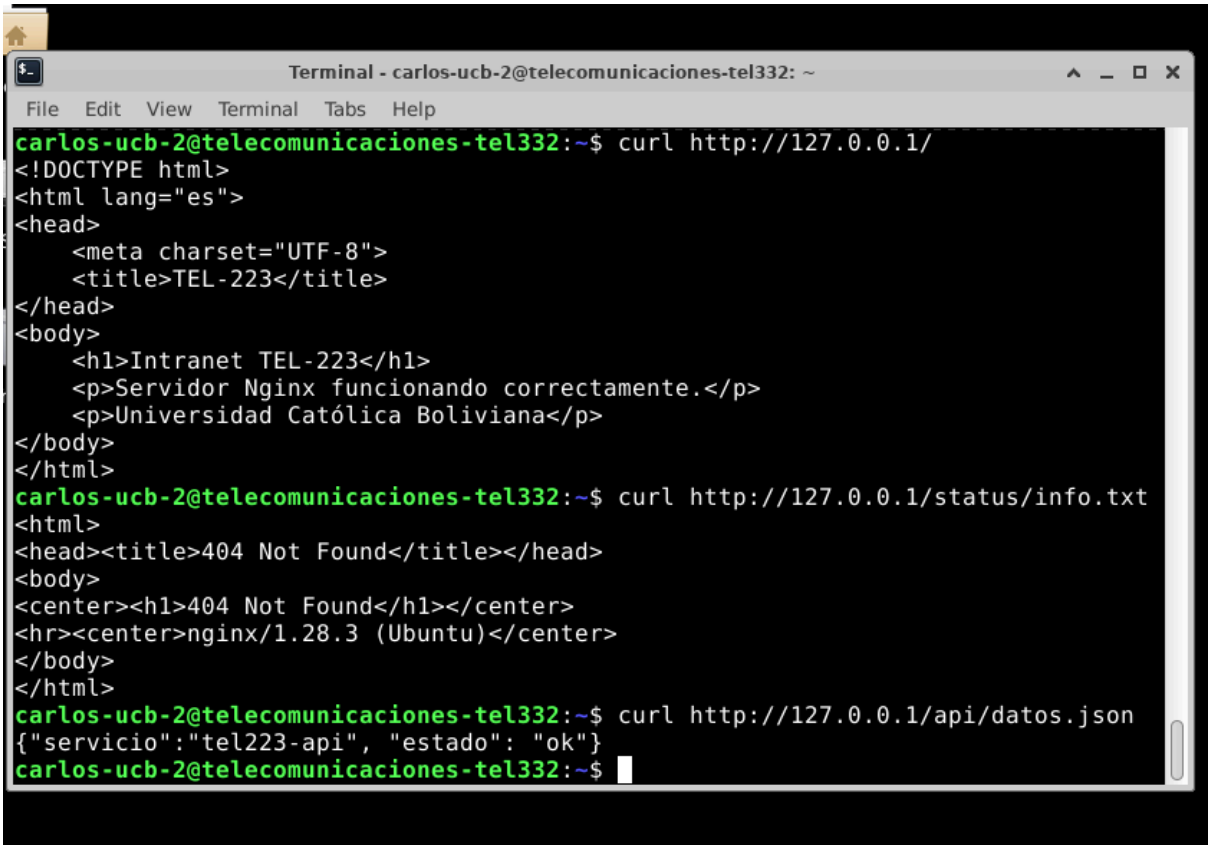


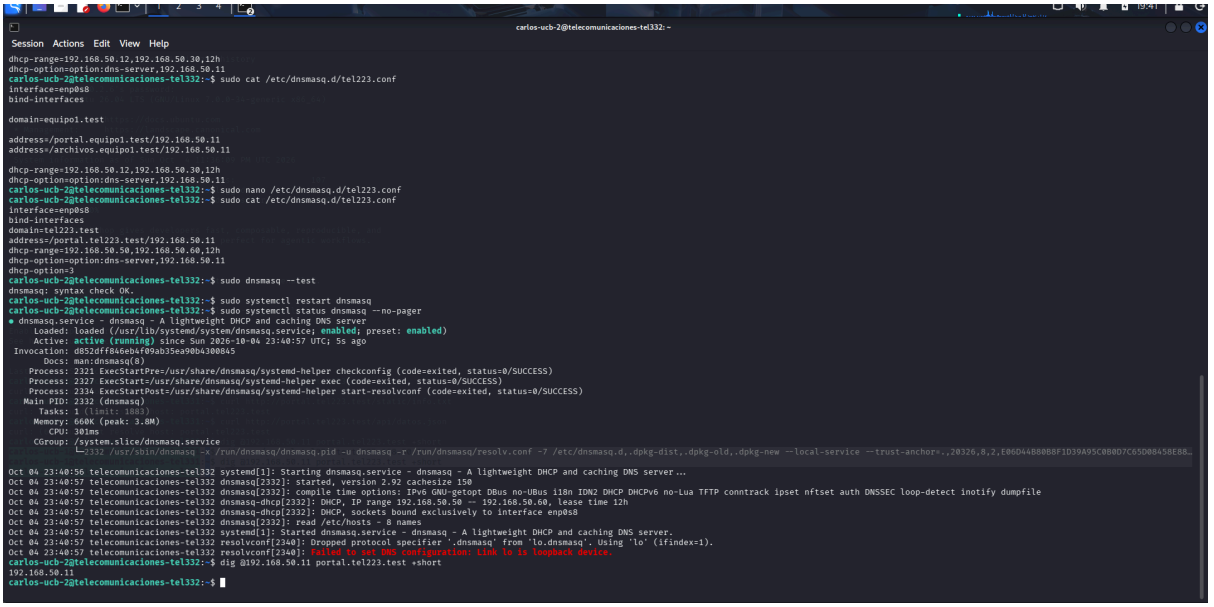


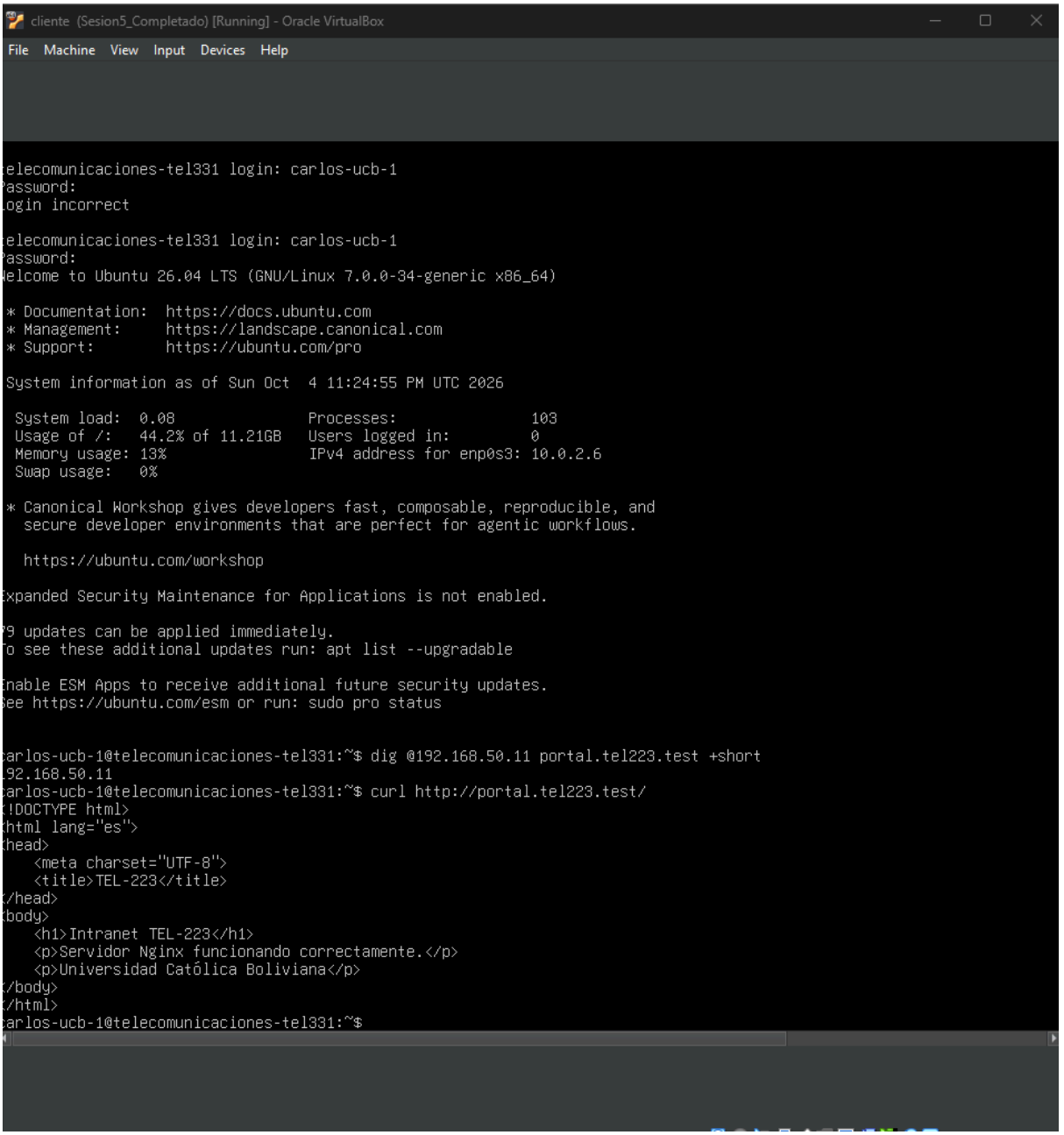

Process: 13936 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_proce
ss on; (code=exited, status=0/SUCCESS)
Process: 13938 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (c
ode=exited, status=0/SUCCESS)
Main PID: 13939 (nginx)
Tasks: 2 (limit: 1883)
Memory: 2.3M (peak: 2.4M)
CPU: 45ms
CGroup: /system.slice/nginx.service
13939 "nginx: master process /usr/sbin/nginx -g daemon on; maste…
13941 "nginx: worker process"
Oct 04 14:04:15 telecomunicaciones-tel332 systemd[1]: Starting nginx.service …..
Oct 04 14:04:15 telecomunicaciones-tel332 systemd[1]: Started nginx.service -…r.
Hint: Some lines were ellipsized, use -l to show in full.
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo ss -ltnp | grep :80
LISTEN 0 5 127.0.0.1:8000 0.0.0.0:* us
ers:(("python3",pid=12260,fd=3))
LISTEN 0 511 0.0.0.0:80 0.0.0.0:* us
ers:(("nginx",pid=13941,fd=5),("nginx",pid=13939,fd=5))
LISTEN 0 511 [::]:80 [::]:* us
ers:(("nginx",pid=13941,fd=6),("nginx",pid=13939,fd=6))
carlos-ucb-2@telecomunicaciones-tel332:~$


Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
Oct 04 14:04:15 telecomunicaciones-tel332 systemd[1]: Starting nginx.service …..
Oct 04 14:04:15 telecomunicaciones-tel332 systemd[1]: Started nginx.service -…r.
Hint: Some lines were ellipsized, use -l to show in full.
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo ss -ltnp | grep :80
LISTEN 0 5 127.0.0.1:8000 0.0.0.0:* us
ers:(("python3",pid=12260,fd=3))
LISTEN 0 511 0.0.0.0:80 0.0.0.0:* us
ers:(("nginx",pid=13941,fd=5),("nginx",pid=13939,fd=5))
LISTEN 0 511 [::]:80 [::]:* us
ers:(("nginx",pid=13941,fd=6),("nginx",pid=13939,fd=6))
carlos-ucb-2@telecomunicaciones-tel332:~$ curl http://127.0.0.1/
<!DOCTYPE html>
<html lang="es">
<head>
<meta charset="UTF-8">
<title>TEL-223</title>
</head>
<body>
<h1>Intranet TEL-223</h1>
Servidor Nginx funcionando correctamente.
Universidad Católica Boliviana
</body>
</html>
carlos-ucb-2@telecomunicaciones-tel332:~$

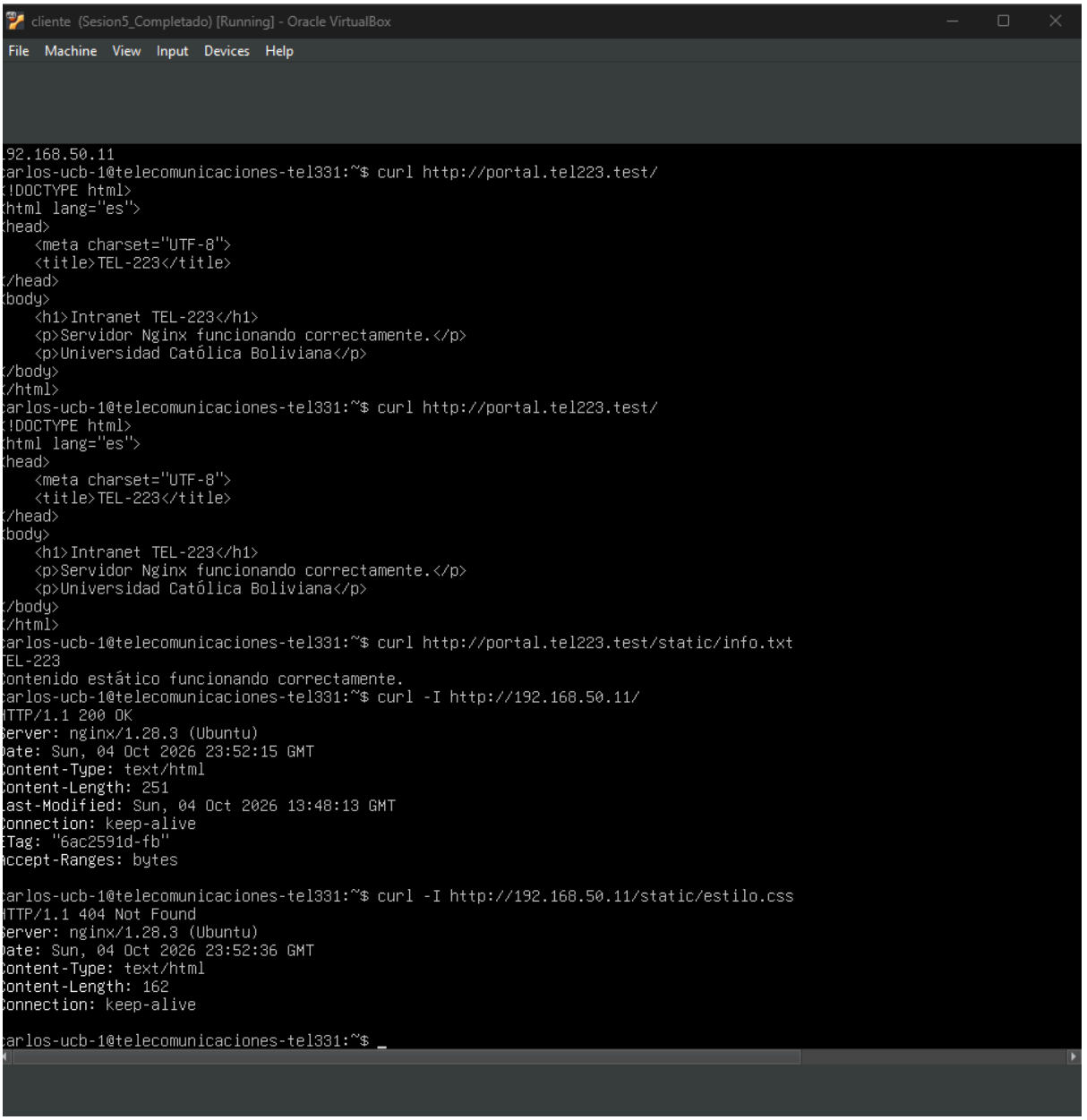

Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
LISTEN 0 511 [::]:80 [::]:* us
ers:(("nginx",pid=13941,fd=6),("nginx",pid=13939,fd=6))
carlos-ucb-2@telecomunicaciones-tel332:~$ curl http://127.0.0.1/
<!DOCTYPE html>
<html lang="es">
<head>
<meta charset="UTF-8">
<title>TEL-223</title>
</head>
<body>
<h1>Intranet TEL-223</h1>
Servidor Nginx funcionando correctamente.
Universidad Católica Boliviana
</body>
</html>
carlos-ucb-2@telecomunicaciones-tel332:~$ curl http://127.0.0.1/status/info.txt
<html>
<head><title>404 Not Found</title></head>
<body>
<h1>404 Not Found</h1>
<hr> nginx/1.28.3 (Ubuntu)
</body>
</html>
carlos-ucb-2@telecomunicaciones-tel332:~$


Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
carlos-ucb-2@telecomunicaciones-tel332:~$ curl http://127.0.0.1/
<!DOCTYPE html>
<html lang="es">
<head>
<meta charset="UTF-8">
<title>TEL-223</title>
</head>
<body>
<h1>Intranet TEL-223</h1>
Servidor Nginx funcionando correctamente.
Universidad Católica Boliviana
</body>
</html>
carlos-ucb-2@telecomunicaciones-tel332:~$ curl http://127.0.0.1/status/info.txt
<html>
<head><title>404 Not Found</title></head>
<body>
<h1>404 Not Found</h1>
<hr> nginx/1.28.3 (Ubuntu)
</body>
</html>
carlos-ucb-2@telecomunicaciones-tel332:~$ curl http://127.0.0.1/api/datos.json
{"servicio":"tel223-api", "estado": "ok"}
carlos-ucb-2@telecomunicaciones-tel332:~$

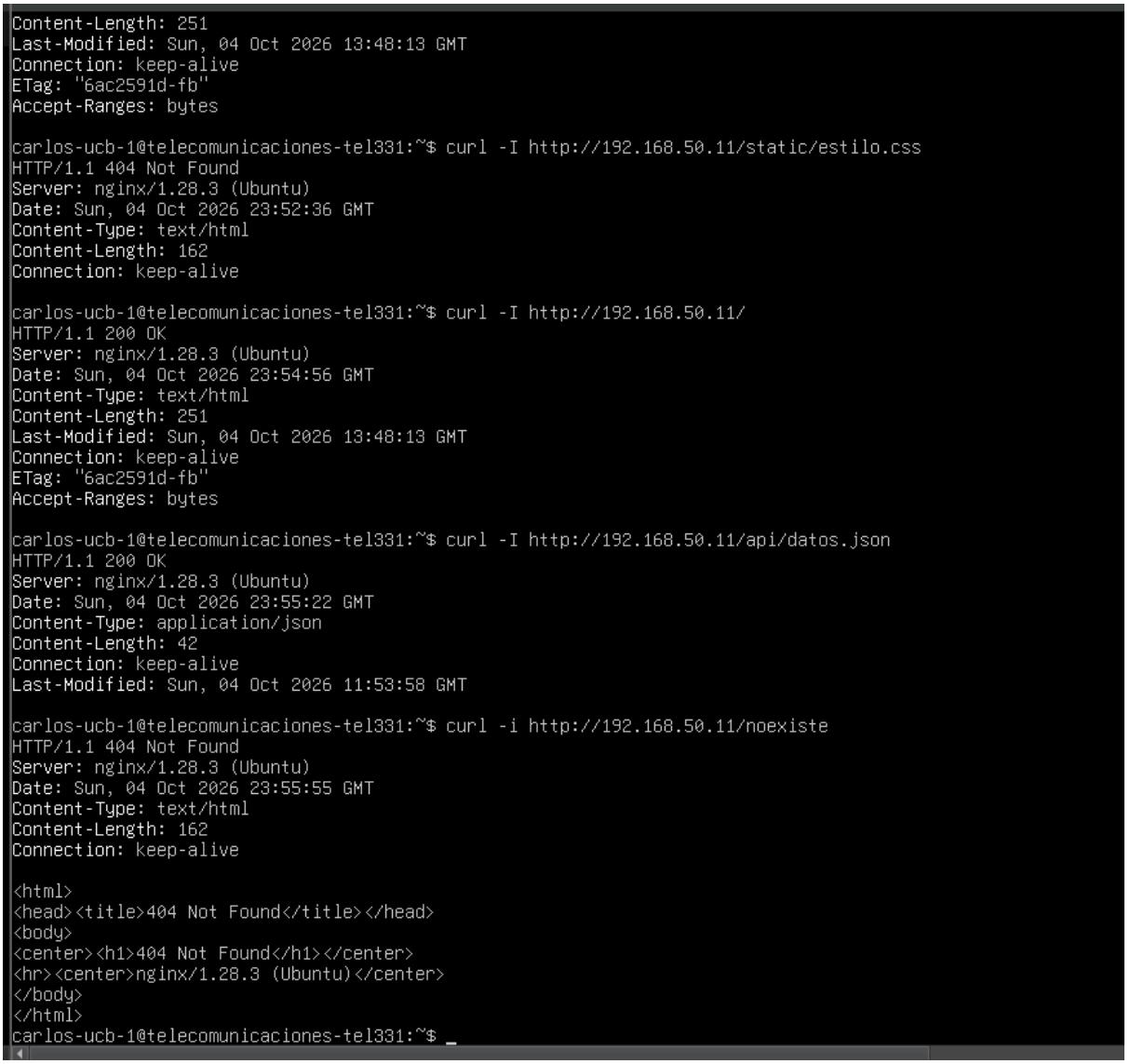

carlos-ucb-2@telecomunicaciones-tel332: ~
Session Actions Edit View Help
dhcp-range=192.168.50.12,192.168.50.30,12h
dhcp-option=option:dns-server,192.168.50.11
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo cat /etc/dnsmasq.d/tel223.conf
interface=enp0s8
bind-interfaces
domain=equipo1.test
address=/portal.equipo1.test/192.168.50.11
address=/archivos.equipo1.test/192.168.50.11
dhcp-range=192.168.50.12,192.168.50.30,12h
dhcp-option=option:dns-server,192.168.50.11
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo nano /etc/dnsmasq.d/tel223.conf
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo cat /etc/dnsmasq.d/tel223.conf
interface=enp0s8
bind-interfaces
domain=tel223.test
address=/portal.tel223.test/192.168.50.11
dhcp-range=192.168.50.50,192.168.50.60,12h
dhcp-option=option:dns-server,192.168.50.11
dhcp-option=3
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo dnsmasq --test
dnsmasq: syntax check OK.
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo systemctl restart dnsmasq
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo systemctl status dnsmasq --no-pager
● dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server
Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service; enabled; preset: enabled)
Active: active (running) since Sun 2026-10-04 23:40:57 UTC; 5s ago
Invocation: d852dff8a6eb4f89ab35ea90b430084
Docs: man:dnsmasq(8)
Process: 2321 ExecStartPre=/usr/share/dnsmasq/systemd-helper checkconfig (code=exited, status=0/SUCCESS)
Process: 2327 ExecStart=/usr/share/dnsmasq/systemd-helper exec (code=exited, status=0/SUCCESS)
Process: 2334 ExecStartPost=/usr/share/dnsmasq/systemd-helper start-resolvconf (code=exited, status=0/SUCCESS)
Main PID: 2332 (dnsmasq)
Tasks: 1 (limit: 1084)
Memory: 660K (peak: 3.8M)
CPU: 301ms
CGroup: /system.slice/dnsmasq.service
└─2332 /usr/sbin/dnsmasq -x /run/dnsmasq/dnsmasq.pid -u dnsmasq -r /run/dnsmasq/resolv.conf -7 /etc/dnsmasq.d,.dpkg-dist,.dpkg-old,.dpkg-new --local-service --trust-anchor=.,20326,8,2,E06D44B80B8F1D39A95C0B0D7C65D08458E88
Oct 04 23:40:56 telecomunicaciones-tel332 systemd[1]: Starting dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server...
Oct 04 23:40:57 telecomunicaciones-tel332 dnsmasq[2332]: started, version 2.92 cachesize 150
Oct 04 23:40:57 telecomunicaciones-tel332 dnsmasq[2332]: compile time options: IPv6 GNU-getopt DBus no-UBus i18n IDN2 DHCP DHCPv6 no-Lua TFTP conntrack ipset nftset auth DNSSEC loop-detect inotify dumpfile
Oct 04 23:40:57 telecomunicaciones-tel332 dnsmasq-dhcp[2332]: DHCP, IP range 192.168.50.50 -- 192.168.50.60, lease time 12h
Oct 04 23:40:57 telecomunicaciones-tel332 dnsmasq-dhcp[2332]: DHCP, sockets bound exclusively to interface enp0s8
Oct 04 23:40:57 telecomunicaciones-tel332 dnsmasq[2332]: read /etc/hosts - 8 names
Oct 04 23:40:57 telecomunicaciones-tel332 systemd[1]: Started dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server.
Oct 04 23:40:57 telecomunicaciones-tel332 resolvconf[2340]: Dropped protocol specifier '.dnsmasq' from 'lo.dnsmasq'. Using 'lo' (ifindex=1).
Oct 04 23:40:57 telecomunicaciones-tel332 resolvconf[2340]: Failed to set DNS configuration: Link lo is loopback device.
carlos-ucb-2@telecomunicaciones-tel332:~$ dig @192.168.50.11 portal.tel223.test +short
192.168.50.11
carlos-ucb-2@telecomunicaciones-tel332:~$



ETAPA 4 — Nginx: contenido estático y proxy inverso



ETAPA 5 — Lectura de transacciones HTTP y registros



Consulta	Código esperado	Content-Type aproximado	Quién responde
/	200	text/html	Nginx
/static/estilo.css	200	text/css	Nginx
/api/datos.json	200	application/json	Backend mediante Nginx
/noexiste	404	text/html	Nginx

```
/html>
arlos-ucb-1@telecomunicaciones-tel331:~$ curl -v http://192.168.50.11/api/datos.json
   Trying 192.168.50.11:80...
 Established connection to 192.168.50.11 (192.168.50.11 port 80) from 192.168.50.25 port 34698
 using HTTP/1.x
 GET /api/datos.json HTTP/1.1
 Host: 192.168.50.11
 User-Agent: curl/8.18.0
 Accept: */*

 Request completely sent off
 HTTP/1.1 200 OK
 Server: nginx/1.28.3 (Ubuntu)
 Date: Sun, 04 Oct 2026 23:57:24 GMT
 Content-Type: application/json
 Content-Length: 42
 Connection: keep-alive
 Last-Modified: Sun, 04 Oct 2026 11:53:58 GMT

"servicio":"tel223-api", "estado": "ok"}
 Connection #0 to host 192.168.50.11:80 left intact
arlos-ucb-1@telecomunicaciones-tel331:~$ _
```

> GET /api/datos.json HTTP/1.1 debajo de using ósea funcional

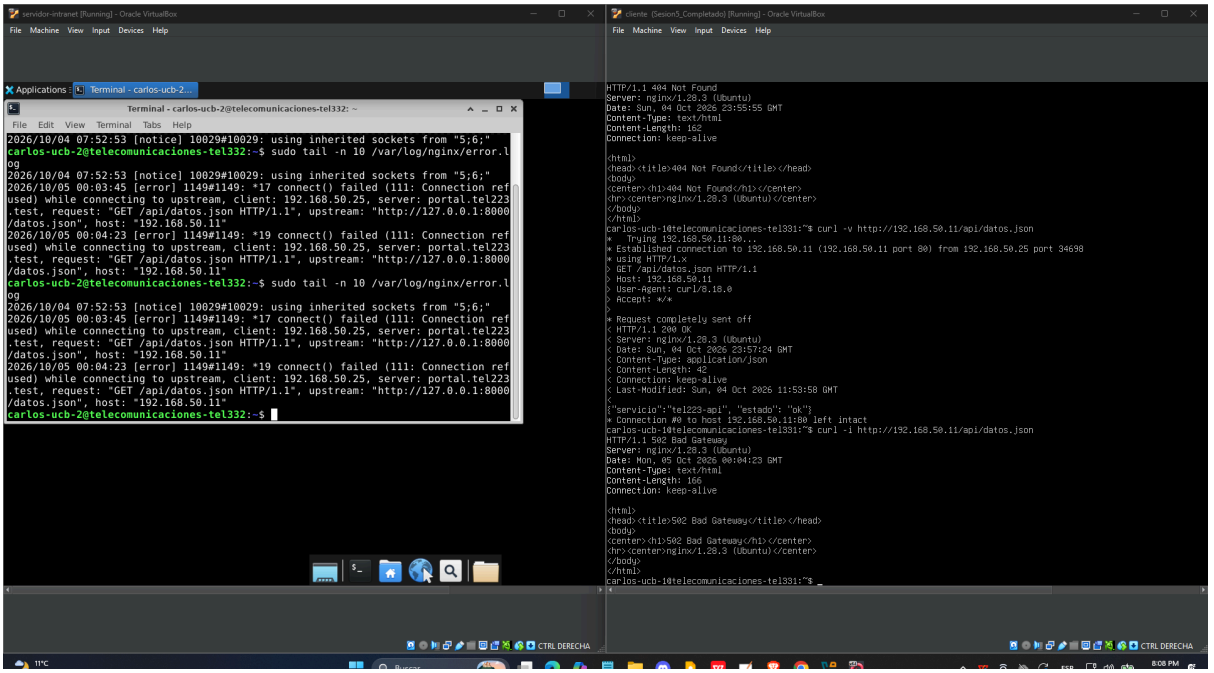
ETAPA 6 — Diagnóstico de fallas con el árbol

```
arlos-ucb-1@telecomunicaciones-tel331:~$ curl -i http://192.168.50.11/api/datos.json
TTP/1.1 502 Bad Gateway
erver: nginx/1.28.3 (Ubuntu)
ate: Mon, 05 Oct 2026 00:04:23 GMT
ontent-Type: text/html
ontent-Length: 166
onnection: keep-alive

html>
head><title>502 Bad Gateway</title></head>
body>
center><h1>502 Bad Gateway</h1></center>
hr><center>nginx/1.28.3 (Ubuntu)</center>
/body>
/html>
arlos-ucb-1@telecomunicaciones-tel331:~$
```

error 502

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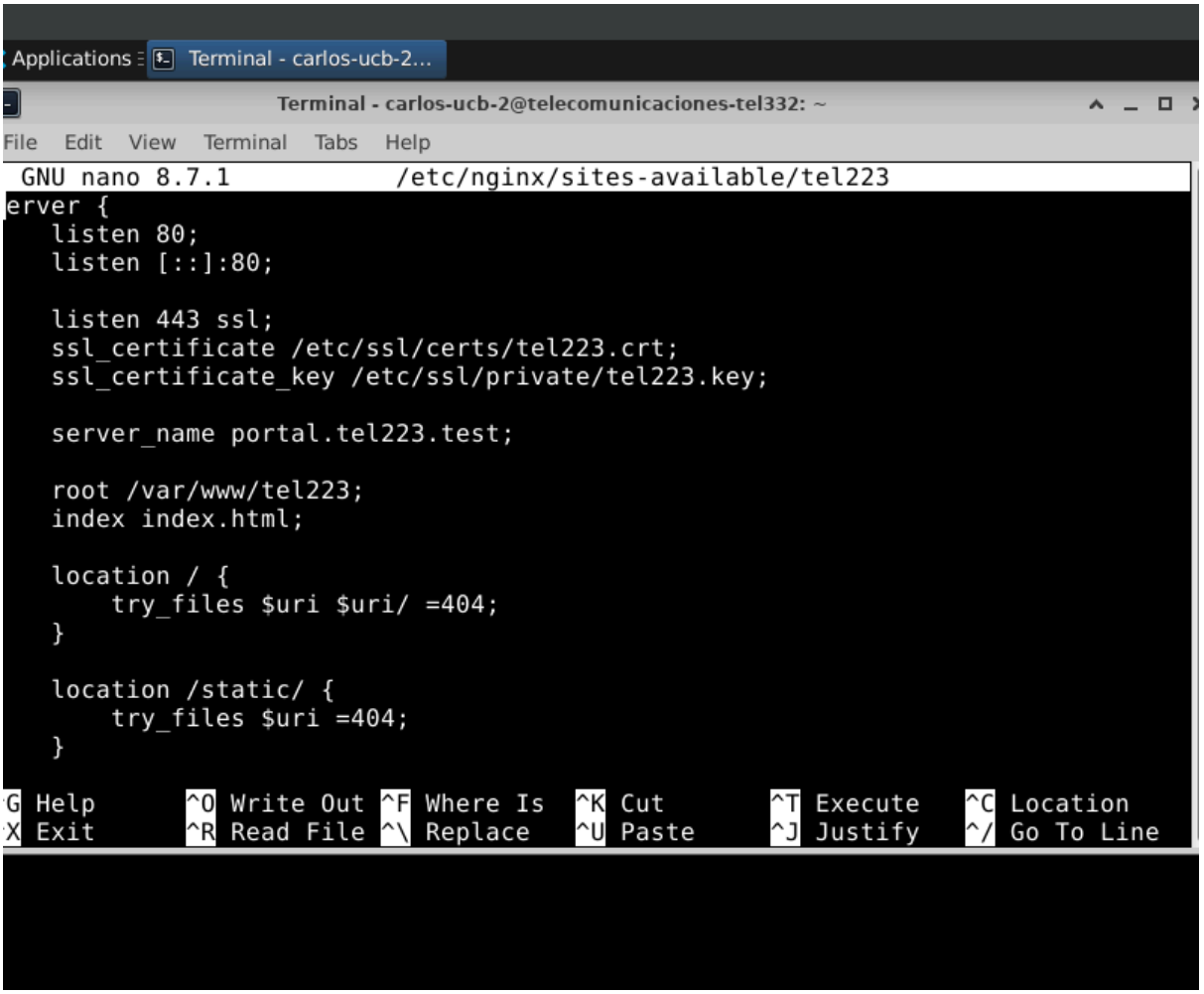


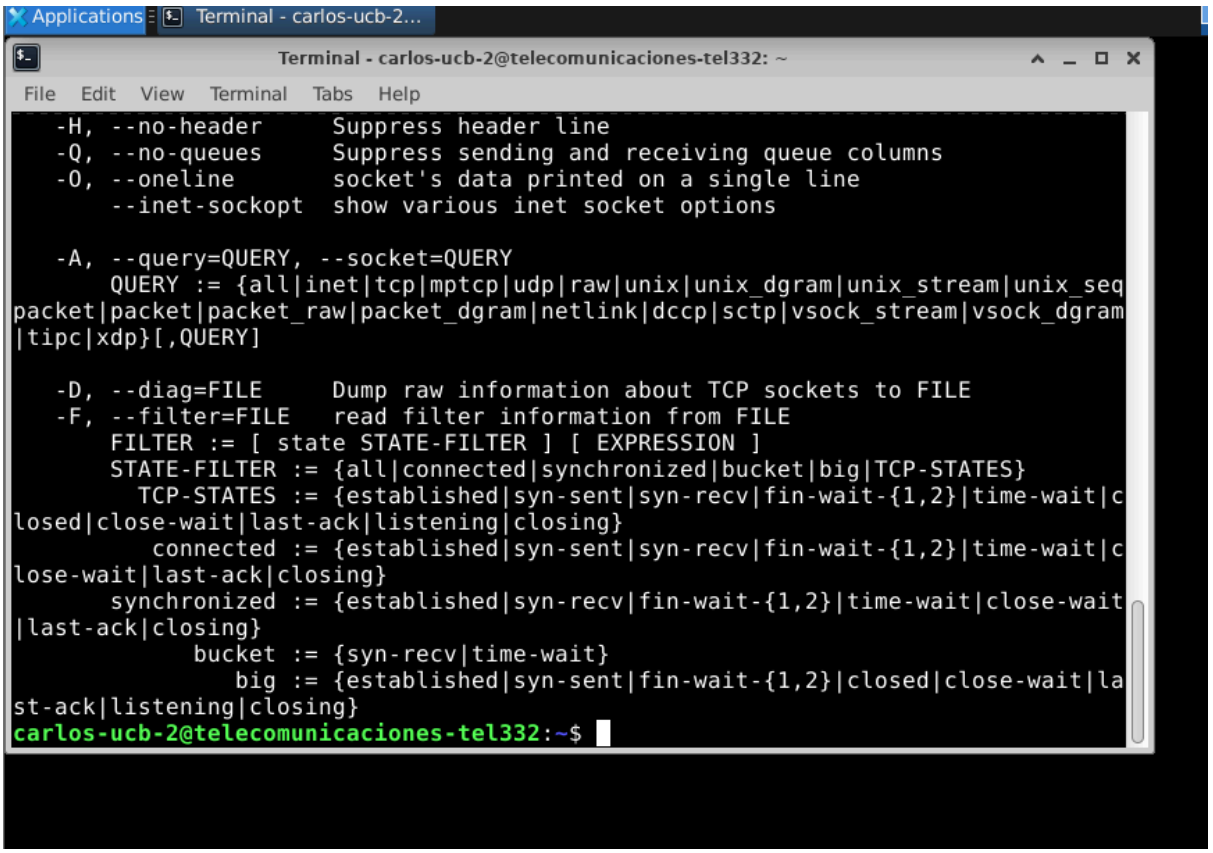
Bitácora de diagnóstico

Falla	Síntoma observado	Primer control que falla	Capa probable	Evidencia (comando y salida)	Causa confirmada
F1	El cliente no puede comunicarse con el servidor	ping 192.168.50.11	Red / conectividad	ping -c 4 192.168.50.11 → pérdida de paquetes o Destination Host Unreachable	Falla de conectividad entre CLIENTE y SERVIDOR
F2	portal.tel223.test no resuelve	dig @192.168.50.11 portal.tel223.test +short	DNS	dig → no devuelve 192.168.50.11	Configuración DNS incorrecta en dnsmasq
F3	El nombre resuelve, pero el sitio no responde	curl http://portal.tel223.test/	HTTP / Nginx	curl → Connection refused o error de conexión	Nginx detenido, puerto 80 no disponible o configuración incorrecta
F4	/api/datos.json devuelve 502 Bad Gateway	curl -i http://192.168.50.11/api/datos.json	Backend / proxy	HTTP/1.1 502 Bad Gateway + connect() failed (111: Connection refused)	Backend tel223-api detenido; Nginx no puede conectar a 127.0.0.1:800

					0
F5	El servicio funciona pero una ruta devuelve 404 Not Found	curl -i http://192.168.50.11/noexiste	Aplicación web / Nginx	HTTP/1.1 404 Not Found	La ruta o recurso solicitado no existe

EXTENSIÓN OPCIONAL — HTTPS con certificado autofirmado


Applications
Terminal - carlos-ucb-2...
Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
GNU nano 8.7.1 /etc/nginx/sites-available/tel223
erver {
listen 80;
listen [::]:80;
listen 443 ssl;
ssl_certificate /etc/ssl/certs/tel223.crt;
ssl_certificate_key /etc/ssl/private/tel223.key;
server_name portal.tel223.test;
root /var/www/tel223;
index index.html;
location / {
try_files $uri $uri/ =404;
}
location /static/ {
try_files $uri =404;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


Applications
Terminal - carlos-ucb-2...
Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
-H, --no-header Suppress header line
-Q, --no-queues Suppress sending and receiving queue columns
-O, --oneline socket's data printed on a single line
--inet-sockopt show various inet socket options
-A, --query=QUERY, --socket=QUERY
QUERY := {all|inet|tcp|mptcp|udp|raw|unix|unix_dgram|unix_stream|unix_seq
packet|packet|packet_raw|packet_dgram|netlink|dccp|sctp|vsock_stream|vsock_dgram
|tipc|xdp}[,QUERY]
-D, --diag=FILE Dump raw information about TCP sockets to FILE
-F, --filter=FILE read filter information from FILE
FILTER := [state STATE-FILTER] [EXPRESSION]
STATE-FILTER := {all|connected|synchronized|bucket|big|TCP-STATES}
TCP-STATES := {established|syn-sent|syn-recv|fin-wait-{1,2}|time-wait|c
losed|close-wait|last-ack|listening|closing}
connected := {established|syn-sent|syn-recv|fin-wait-{1,2}|time-wait|c
lose-wait|last-ack|closing}
synchronized := {established|syn-recv|fin-wait-{1,2}|time-wait|close-wait
|last-ack|closing}
bucket := {syn-recv|time-wait}
big := {established|syn-sent|fin-wait-{1,2}|closed|close-wait|la
st-ack|listening|closing}
carlos-ucb-2@telecomunicaciones-tel332:~$

```
body>
center><h1>502 Bad Gateway</h1></center>
hr><center>nginx/1.28.3 (Ubuntu)</center>
/body>
/html>
arlos-ucb-1@telecomunicaciones-tel331:~$ curl -I http://192.168.50.11/
TTP/1.1 200 OK
erver: nginx/1.28.3 (Ubuntu)
ate: Mon, 05 Oct 2026 00:22:04 GMT
ontent-Type: text/html
ontent-Length: 251
ast-Modified: Sun, 04 Oct 2026 13:48:13 GMT
onnection: keep-alive
Tag: "6ac2591d-fb"
ccept-Ranges: bytes

arlos-ucb-1@telecomunicaciones-tel331:~$ curl -k -I https://192.168.50.11/
TTP/1.1 200 OK
erver: nginx/1.28.3 (Ubuntu)
ate: Mon, 05 Oct 2026 00:22:31 GMT
ontent-Type: text/html
ontent-Length: 251
ast-Modified: Sun, 04 Oct 2026 13:48:13 GMT
onnection: keep-alive
Tag: "6ac2591d-fb"
ccept-Ranges: bytes

arlos-ucb-1@telecomunicaciones-tel331:~$ curl -I http://192.168.50.11/
TTP/1.1 200 OK
erver: nginx/1.28.3 (Ubuntu)
ate: Mon, 05 Oct 2026 00:23:00 GMT
ontent-Type: text/html
ontent-Length: 251
ast-Modified: Sun, 04 Oct 2026 13:48:13 GMT
onnection: keep-alive
Tag: "6ac2591d-fb"
ccept-Ranges: bytes

arlos-ucb-1@telecomunicaciones-tel331:~$ curl -k -I http://192.168.50.11/
TTP/1.1 200 OK
erver: nginx/1.28.3 (Ubuntu)
ate: Mon, 05 Oct 2026 00:23:22 GMT
ontent-Type: text/html
ontent-Length: 251
ast-Modified: Sun, 04 Oct 2026 13:48:13 GMT
onnection: keep-alive
Tag: "6ac2591d-fb"
ccept-Ranges: bytes

arlos-ucb-1@telecomunicaciones-tel331:~$
```

Pregunta de repaso	Mi respuesta
curl indica self-signed certificate. ¿Por qué eso impide confiar en el servidor?	Porque el certificado fue firmado por el propio servidor y no por una autoridad certificadora (CA) reconocida por el cliente. Por eso, el cliente no puede verificar una cadena de confianza que confirme la identidad del servidor.
Aunque confiara en el certificado, ¿qué otro problema tendría al pedirlo por IP? (Pista: compare el CN del certificado con lo que escribió en curl.)	El certificado fue creado para `portal.tel223.test` (`CN=portal.tel223.test`), pero la solicitud se realizó a `https://192.168.50.11/`. La IP no coincide con el nombre para el que fue emitido el certificado, por lo que existiría un error de coincidencia de identidad.
¿Hasta qué punto protege TLS en esta arquitectura?	TLS cifra y protege la comunicación entre el cliente y Nginx, proporcionando confidencialidad

	e integridad durante el transporte. Sin embargo, el certificado autofirmado no permite verificar de forma confiable la identidad del servidor. Además, TLS no corrige problemas de DNS, Nginx o del backend.